



UNIX SHELL

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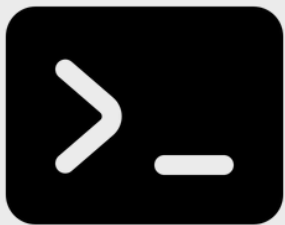
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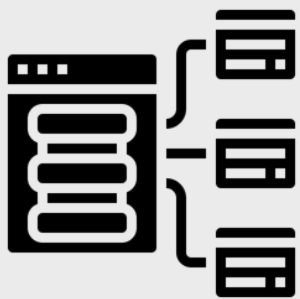
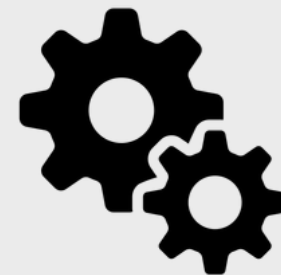


OVERVIEW



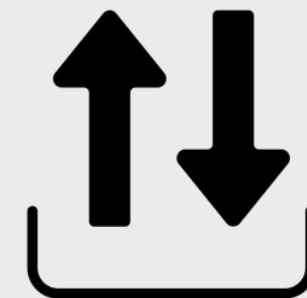
**UNIX like command
shell ton top of OS**

**Execute processes
independently**



**File descriptor
abstractor**

**read/write
interface**



MOTIVATION



Batch System

- Non-interactive
- One job at a time
- Cannot chain commands
- Hard to automate



UNIX System

- Interactive shell
- Multi-tasking
- Commands combined with pipes
- Shell scripting and process automation



CORE PROBLEMS

- ✓ **Lack of interactive command execution**
 - ✓ **Absence of inter-process communication**
 - ✓ **Multiple I/O mechanisms without abstraction**
 - ✓ **Difficulty in automating command workflows**
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SOLUTION APPROACH

- ✓ **Independent process execution**
 - ✓ **Pipe-based inter-process communication**
 - ✓ **Unified file descriptor I/O**
 - ✓ **Command chaining and redirection**
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OBJECTIVE

- **IMPLEMENT SHELL COMMANDS AS INDEPENDENT USER PROCESSES**
 - **MANAGE PROCESS CREATION AND SYNCHRONIZATION**
 - **UNIFY CONSOLE I/O, IPC, AND FILE SYSTEM OPERATIONS USING READ/WRITE**
 - **SUPPORT PIPELINES AND INPUT/OUTPUT REDIRECTION**
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RELEVANT OS CONCEPTS



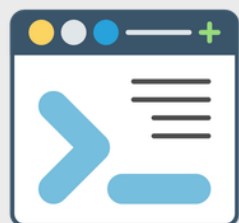
Process Management

File Descriptor Abstraction



Read/Write Operation

Inter Process Communication



Kernel - Shell Interaction



THANK YOU

